Practice Set 13 Paper 2 – Urology

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Practice paper compiled from the question bank.

Q1. Describe role of nephron sparing surgery in management of RCC

Type: LAQ (25 marks) (also answered in 2022-Summer Paper 3, Q1 and 2025-Winter Paper 1, Q2)

Definition and rationale

- NSS (nephron-sparing surgery) is complete removal of the tumour with **maximal preservation of functioning parenchyma**. PN (partial nephrectomy) is the standard form; thermal ablation is the non-excisional form.
- **Why it matters:** RN (radical nephrectomy) can cause CKD (chronic kidney disease), linked to cardiovascular events and death. PN keeps about **90%** of global function (RN 65–70%) with equal cancer control for small masses.
- **CKD from surgery alone** is stable, with near-normal survival if the new GFR (glomerular filtration rate) stays >45; **CKD from medical disease** (diabetes, hypertension) progresses. So NSS matters most when medical renal disease coexists.

Role by clinical setting

Setting	Role of NSS
cT1a (≤4 cm)	PN is the **standard of care** when intervention is needed; RN is overtreatment
cT1b–T2	PN when feasible and function matters; otherwise debatable
Hereditary RCC (renal cell carcinoma)	Observe until the largest tumour is **3 cm**, then enucleate (except HLRCC [hereditary leiomyomatosis and RCC]: wide excision)
Bilateral tumours, solitary kidney	Imperative PN, staged if bilateral; avoids dialysis
Frail, comorbid, cT1a <3 cm	Thermal ablation or active surveillance
Locally advanced, thrombus, cN1, metastatic	Generally RN or cytoreductive surgery

Indications

Category	Situations
Imperative	Bilateral RCC; solitary functioning kidney; RN would mean dialysis
Relative	Contralateral kidney threatened: stones, renal artery stenosis, diabetes, hypertension, proteinuria; hereditary RCC
Elective	Localised tumour with a normal contralateral kidney

- **AUA (American Urological Association) 2021:** RN is preferred only if **all three** apply: high complexity; no CKD or proteinuria; normal contralateral kidney with new GFR likely **>45** after RN. Otherwise PN.
- **Relative contraindications: <20% of nephron mass retained**; pedicle encasement; venous tumour thrombus; cT4.

Preoperative evaluation

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging); chest imaging; eGFR (estimated GFR) and urine protein.
- **R.E.N.A.L. nephrometry** (Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location to polar lines): **4–6 low, 7–9 moderate, 10–12 high** complexity. PADUA is an alternative.

- **Biopsy** when it changes management (about **20%** of enhancing cT1 masses are benign). Genetic counselling if age ≤46, multifocal or bilateral.

Techniques of PN

Approach: open (flank extraperitoneal), laparoscopic, or **RAPN (robot-assisted PN)**: transperitoneal for anterior and retroperitoneal for posterior tumours.

Steps

1. **Early hilar control** of artery, vein and accessory vessels, even if off-clamp is planned.
2. Defat the kidney; **intraoperative ultrasound** maps borders and depth.
3. Clamp the artery. Use **ice slush** if ischaemia is expected to exceed **25–30 minutes** (solitary kidney, CKD).
4. Excise with a thin rim. **A negative margin matters; its width does not.**
5. **Two-layer renorrhaphy:** inner layer closes vessels and collecting system; outer capsular layer over a haemostatic bolster.
6. Unclamp; check haemostasis; close Gerota fascia; drain.

	Standard (wedge) resection	Enucleation
Plane	Through normal parenchyma	Along the pseudocapsule
Advantage	Fewer positive margins	Less bleeding, maximal parenchyma
Use	Default	Hereditary, multifocal, severe CKD

- **Ischaemia:** warm ischaemia **<25–30 minutes**. Function depends mainly on the **parenchyma preserved**; off-clamp, selective clamping and mannitol show no long-term eGFR benefit.

Thermal ablation

- **Cryoablation** (double freeze–thaw) or **RFA (radiofrequency ablation)**; percutaneous preferred.
- AUA: alternative for **cT1a <3 cm**; biopsy before (preferred) or at ablation. Local recurrence is higher than PN but treatable by repeat ablation.
- **Also know:** MWA (microwave ablation), SBRT (stereotactic body radiotherapy) for non-surgical candidates.

Outcomes and evidence

- **Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes. RAPN: shorter ischaemia than laparoscopy, less blood loss than open.
- Elective PN: local recurrence **1–2%**; imperative 3–5%.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** RN vs PN for ≤5 cm tumours: PN had better function and similar oncology; 10-year overall survival 81% RN vs 76% PN. The trial is flawed.

Complications

- **Urine leak** (drain, stent); **haemorrhage, pseudoaneurysm or AV (arteriovenous) fistula** (delayed haematuria: angioembolisation); acute kidney injury; temporary dialysis in a solitary kidney.
- **Positive margin 0–7%:** surveillance, not reoperation.
- **Hyperfiltration injury** when >50% of a solitary kidney is removed: proteinuria, FSGS (focal segmental glomerulosclerosis); give an ACE (angiotensin-converting enzyme) inhibitor.
- **Follow-up:** risk-based imaging; eGFR and urine protein.

Recent advances

- 3D (three-dimensional) models; ICG (indocyanine green) for selective clamping; neoadjuvant TKI (tyrosine kinase inhibitor) downsizing in imperative cases.

Diagram to draw: clamped renal artery, tumour excised with a rim, two-layer renorrhaphy.

> **Key points to remember:**
>
> - PN is the standard for cT1a; ablation or surveillance for frail patients with tumours <3 cm.
> - AUA: RN only if complex + no CKD + new GFR likely >45.
> - Preserved parenchyma, not clamp time, governs function.
> - EORTC 30904: better function, similar cancer control.

Q2. Desc. factors influencing outcome of ESWL. Discuss complic. & strategies to avoid them.

Type: LAQ (25 marks) (also answered in 2021-Summer Paper 2, Q1; mechanism and indications in 2019-Summer Paper 2, Q2)

Introduction

- ESWL (extracorporeal shock wave lithotripsy, now SWL) breaks stones with shock waves (peak 30–100 MPa, negative tail about −10 MPa) generated at F1 and focused on the stone at F2.
- Stones break by spallation, shear, squeezing and **cavitation**. Cavitation also causes most tissue injury, so steps that improve outcome usually reduce complications too.
- Outcome = SFR (stone-free rate) and need for retreatment.

Factors influencing outcome

1. Stone factors

Factor	Favourable	Unfavourable
Size (most important)	<10 mm	Renal >20 mm (PCNL [percutaneous nephrolithotomy] first); ureteric ≥10 mm favours URS (ureteroscopy)
Density on NCCT (non-contrast CT [computed tomography])	<900 HU (Hounsfield units)	>1000 HU
Composition	Calcium oxalate dihydrate, uric acid	Cystine, brushite, calcium oxalate monohydrate
Impaction (ureter)	Recent, mobile	Chronic impaction: thick ureteral wall around the stone on CT
Burden and site	Single, renal pelvis or upper calyx	Multiple sites; lower pole >1 cm

2. Renal anatomical factors

- **Lower pole:** fragments drain against gravity. **Lower Pole I (Albala, 2001):** stones >1 cm, SFR **21% SWL vs 91% PCNL**. A steep infundibulopelvic angle and a long, narrow infundibulum reduce clearance.
- Obstruction distal to the stone (UPJO [ureteropelvic junction obstruction], infundibular stenosis, calyceal diverticulum) traps fragments. Horseshoe and ectopic kidneys drain poorly.

3. Patient factors

- **SSD (skin-to-stone distance) >10 cm** (obesity) predicts failure. Skeletal deformity can prevent targeting.
- Respiratory movement of 10–15 mm moves the stone out of F2.
- **Triple D score:** stone density, stone volume and skin-to-stone distance together predict success.

4. Machine and technique factors

- **Energy source:** electrohydraulic (HM3) breaks best; piezoelectric has low energy and a tiny focal zone; modern machines are electromagnetic. A focal zone wider than the stone breaks better.
- **Coupling:** air pockets in the gel cut fragmentation by up to two-fifths.
- **Rate 60–90 shocks/min** beats 120/min, because the bubble cloud clears between shocks. **Ramping** energy up also helps.

- **Analgesia:** general anaesthesia or sedation keeps the stone in focus.
- Prone position for mid-ureteric stones over the sacrum; continuous imaging; experienced team. Little gain beyond two sessions.
- **Adjuncts:** α-blockers (faster clearance); potassium citrate; percussion, diuresis and inversion for lower pole fragments.

Complications

Group	Complication	Comment
Fragment-related	Renal colic; **steinstrasse** ("stone street" of fragments in the ureter); residual fragments	Steinstrasse rises several-fold with burden >2 cm^2
Infective	UTI (urinary tract infection), pyelonephritis, urosepsis	Infection stones, stents
Renal tissue injury	Transient haematuria (common); **subcapsular or perirenal haematoma** (minority)	Risk: age, hypertension, obesity, diabetes, coronary disease, anticoagulation, high energy, many shocks
Other organs	Liver, spleen, pancreas, colon, lung (children)	Very rare
Cardiac	Arrhythmia during treatment	ECG (electrocardiogram) gating
Skin	Petechiae, bruising	Minor

- **Refuted long-term effects:** new hypertension, diabetes, loss of renal function, infertility.

Strategies to avoid complications

1. **Select.** Absolute contraindications: pregnancy; uncorrected coagulopathy or ongoing antithrombotics; untreated UTI; obstruction distal to the stone; aneurysm near the stone (aortic >4 cm); severe untreated hypertension. Send large, dense, lower pole or impacted stones to URS or PCNL.
2. **Prepare.** Urine culture; treat bacteriuria. Prophylaxis only with risk factors (stent, nephrostomy, struvite, recent UTI). Stop antithrombotics before and for about 48 hours after. Control blood pressure; check pacemakers.
3. **Protect the kidney.** Slow rate; **slow ramping** (induces protective renal vasoconstriction and lowers haematoma risk); fewest shocks and lowest effective energy.
4. **Prevent steinstrasse.** PCNL or URS for large burdens. Routine stenting does not improve SFR, though it may reduce steinstrasse with large stones. α-blockers after SWL.
5. **Treat early.** Steinstrasse: MET (medical expulsive therapy), SWL to the lead fragment or URS; **nephrostomy if fever or obstruction**. Haematoma: bed rest, transfusion, embolisation if bleeding continues.
6. **Follow up** with KUB (kidney-ureter-bladder radiograph) and ultrasound.

Recent advances

- **Burst wave lithotripsy** (ultrasound bursts: less cavitation, no anaesthesia, uniform fragments) and **ultrasonic propulsion** of fragments.

Diagram to draw: ellipsoid reflector with F1, F2, stone and gel coupling; box of failure predictors.

Key points to remember:

- Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
- Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
- Slow rate (60–90/min), ramping and good coupling improve breakage and reduce injury.
- Steinstrasse risk rises with burden >2 cm^2; no routine stent.

Q3a. Bicalutamide

Type: SAQ (10 marks) (combined androgen blockade is answered in 2018-Summer Paper 4, Q3f)

Introduction and mechanism

- Non-steroidal **AR (androgen receptor) antagonist**, called first- or second-generation. It competes with testosterone and DHT (dihydrotestosterone) at the ligand-binding domain.
- Given alone, it blocks central feedback, so **LH (luteinising hormone) and testosterone rise** (about 1.5 times); aromatisation to oestradiol causes gynaecomastia; libido is partly preserved.
- **Half-life about 6 days** (once daily); unaffected by age or renal impairment. Most potent and best tolerated of its class.

Uses

Use	Dose	Evidence
CAB (combined androgen blockade) with LHRH (LH-releasing hormone) agonist, metastatic	50 mg daily	Small survival gain (2–3%, PCTCG [Prostate Cancer Trialists' Collaborative Group])
Flare prevention	50 mg, with or before the agonist, 2–4 weeks	Main use today
Monotherapy, locally advanced	150 mg daily	**EPC (Early Prostate Cancer) programme:** better progression-free survival; **harm in localised disease** (death HR [hazard ratio] 1.23)
With salvage radiotherapy	150 mg for 2 years	**RTOG 9601:** 12-year overall survival 76.3% vs 71.3%
CRPC (castration-resistant prostate cancer)	50 mg	Second-line step; largely replaced by newer agents

- 150 mg monotherapy roughly equals castration in locally advanced disease, with better sexual interest; not FDA (US Food and Drug Administration) approved alone, but used in some countries.
- **Antiandrogen withdrawal:** stopping bicalutamide gives a PSA (prostate-specific antigen) fall >50% in **15–30%** for 3.5–5 months. Mutant AR turns the drug into an agonist.
- **Also know:** stuttering priapism; feminising hormone therapy; studied in BPH (benign prostatic hyperplasia).

Adverse effects

- **Gynaecomastia and breast pain** (two-thirds at 150 mg): prevent with breast radiotherapy (10–15 Gy single fraction) or tamoxifen 10 mg daily.
- **Hepatotoxicity** (monitor liver tests); rare **interstitial pneumonitis**; erectile dysfunction; fewer hot flushes than castration.
- **Comparison:** enzalutamide beat bicalutamide in CRPC (TERRAIN, STRIVE) and in **ENZAMET**.

Diagram to draw: hypothalamus–pituitary–testis axis with bicalutamide blocking the AR.

Key points to remember:

- Half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
- Monotherapy raises testosterone; gynaecomastia two-thirds.
- EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
- Withdrawal response 15–30%.

Q3b. Mullerionosis of UB

Type: SAQ (10 marks)

"Mullerionosis of UB" is read as müllerianosis of the urinary bladder.

Definition

- Benign lesions of **müllerian-type epithelium** in the bladder wall, in three forms:

Lesion	Tissue
Endometriosis (commonest)	Endometrial glands **and stroma**
Endocervicosis	Endocervical mucinous glands
Endosalpingiosis	Ciliated tubal epithelium

- **Müllerianosis** strictly means two or more of these together.
- Women of reproductive age, often after **caesarean section**. Usually **posterior wall or dome** (vesicouterine pouch).
- Bladder is the commonest urinary tract site of endometriosis, ahead of the ureter.

Pathogenesis

- **Implantation:** retrograde menstruation (Sampson) into the vesicouterine pouch.
- **Iatrogenic implantation** at caesarean or pelvic surgery (primary vs iatrogenic forms).
- **Metaplasia** of coelomic or subperitoneal müllerian remnants (the "secondary müllerian system").
- Lymphatic or vascular spread; direct extension of anterior uterine adenomyosis.

Clinical features

- **Cyclical haematuria** or irritative LUTS (lower urinary tract symptoms) with menses; suprapubic pain, dysmenorrhoea, dyspareunia.
- Endocervicosis: non-cyclical mass or LUTS.
- **Also know:** menouria differential includes vesicouterine fistula (Youssef syndrome).

Evaluation

- **Ultrasound** (transabdominal plus transvaginal): size and depth of a hypoechoic nodule.
- **MRI (magnetic resonance imaging):** extent; haemorrhagic foci **bright on T1**, fibrotic thickening dark on T2.
- **Cystoscopy:** bluish submucosal nodule with neovascularity, best seen during menses. Of **limited value**, because disease is mainly **extravesical** (from serosa inwards).
- **Biopsy:** excludes adenocarcinoma, especially with endocervicosis. Immunohistochemistry: ER/PR (oestrogen/progesterone receptor) positive; CD10 marks endometrial stroma.
- Ultrasound of upper tracts for ureteric involvement.
- **Differential:** adenocarcinoma, cystitis glandularis, nephrogenic adenoma, urachal remnant.

Management

- **Hormonal suppression** (combined pill, progestins, GnRH [gonadotropin-releasing hormone] agonists): symptom relief; recurrence after stopping.
- **TUR (transurethral resection):** incomplete for transmural disease; perforation risk; recurs.
- **Partial cystectomy** (laparoscopic or robotic) with a margin: **primary treatment**. Ureteric stent or reimplantation if the orifice is involved.

- Excise other pelvic endometriosis with the gynaecologist.
- Malignant change (endometrioid or clear cell adenocarcinoma) is rare.

Diagram to draw: sagittal pelvis with a dome nodule growing from the vesicouterine pouch into the detrusor.

Key points to remember:

- Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
- Cyclical haematuria; MRI bright on T1.
- Disease is extravesical: cystoscopy limited, partial cystectomy preferred.

Q3c. Brainstem death

Type: SAQ (10 marks) (also answered in 2022-Summer Paper 4, Q3b and 2024-Winter Paper 2, Q3d)

Definition and law

- **Irreversible loss of the capacity for consciousness and breathing**; legally death in India: THOA (Transplantation of Human Organs Act) 1994, amended 2011, Rules 2014.

Pathophysiology

- Raised ICP (intracranial pressure) → coning → Cushing reflex → **catecholamine storm** → loss of sympathetic tone → **hypotension**.
- Consequences: **diabetes insipidus** (polyuria, hypernatraemia), hypothermia, DIC (disseminated intravascular coagulation), neurogenic pulmonary oedema.

Preconditions

- Apnoeic coma on a ventilator with a **known irreversible cause**.
- Exclude sedatives, relaxants, hypothermia, metabolic or endocrine causes, shock.

Tests: all brainstem reflexes absent

Reflex	Method	Cranial nerves
Pupillary	Bright light: pupils fixed	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic	Turn head: eyes move with head	VIII, III, VI
Oculovestibular	Ice-cold water in each ear: no eye movement	VIII, III, VI
Motor in cranial nerve area	Supraorbital pressure: no grimace	V, VII
Gag and cough	Pharynx touch, tracheal suction: no response	IX, X

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect with oxygen insufflation; **no breathing at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg**. Abort for hypotension, desaturation or arrhythmia.
- **Ancillary tests** if examination is impossible: angiography, perfusion scan, EEG (electroencephalogram).

Indian Board of Medical Experts

- Four members: hospital in-charge; independent specialist nominated from an approved panel; neurologist or neurosurgeon; treating doctor. **No transplant-team member.**
- **Two sets of tests at least 6 hours apart**; recorded on **Form 10**.

Clinical implications

- **Mandatory request** for donation; next-of-kin consent; NOTTO (National Organ and Tissue Transplant Organisation) allocates.
- **Donor management, rule of 100s:** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L. Vasopressin, thyroid hormone, steroids, warming.

Diagram to draw: flowchart: coma → preconditions → test 1 → 6 h → test 2 → Form 10 → consent → NOTTO.

> **Key points to remember:**
>
> - Six brainstem reflexes absent; apnoea at $PaCO_2$ ≥60 mmHg.
> - Four-member Board; no transplant-team member.
> - Two tests ≥6 h apart; Form 10.

Q3d. Screening of CA prostate.

Type: SAQ (10 marks)

Concept

- Screening tests healthy, asymptomatic men to cut deaths through earlier cure. The price is **overdiagnosis** (cancers that would never cause harm) and **overtreatment**.
- Since PSA (prostate-specific antigen) testing, US mortality has fallen over 40%.

Tools

- **PSA:** biopsy threshold 4 ng/mL (many use 2.5–3; ERSPC used 3). Repeat a raised PSA before biopsy.
- **DRE (digital rectal examination):** little value alone; abnormal DRE raises risk.
- **Refinements:** PSA density ≥0.15; free PSA ≤25%; PHI (Prostate Health Index), 4Kscore, Stockholm3; urine tests (PCA3 [prostate cancer antigen 3], ExoDx, SelectMDx).
- **MRI (magnetic resonance imaging) before biopsy:** fewer biopsies and less low-grade overdiagnosis.

Evidence

Trial	Result
ERSPC (European Randomised Study of Screening for Prostate Cancer)	About **20% fall in prostate cancer death** at 16 years; 570 invited and 18 extra diagnoses per death prevented
PLCO (Prostate, Lung, Colorectal and Ovarian)	No difference: control arm heavily PSA-tested
Göteborg-1	Larger mortality reduction with biennial screening
Göteborg-2 (2022)	MRI-targeted biopsy only: halved overdiagnosis

- Corrected for non-attendance and contamination, benefit is 27–31%.

Guidelines

Body	Recommendation
USPSTF (US Preventive Services Task Force) 2018	**Grade C**: shared decision at 55–69; not ≥70 (was Grade D in 2012)
AUA (American Urological Association)	Shared decision 55–69; 2-yearly interval
NCCN (National Comprehensive Cancer Network)	Baseline PSA and discussion at 45
EAU (European Association of Urology)	Risk-adapted: informed men from 50, from 45 with family history or African descent, from 40 with BRCA2

- **No body advises unselected population screening.**
- **High risk:** Black men, father or brother diagnosed <65, BRCA2 and other DNA-repair mutations: start earlier.
- Stop when life expectancy is under 10–15 years.

Reducing harm

- **Active surveillance** for Grade Group 1; MRI and biomarkers before biopsy; intervals by baseline PSA.

Diagram to draw: pathway: PSA → repeat → risk calculator or MRI → targeted biopsy → active surveillance or treatment.

Key points to remember:

- ERSPC: about 20% fall in prostate cancer death; PLCO null from contamination.
- USPSTF Grade C at 55–69.
- MRI-first and active surveillance reduce overdiagnosis and overtreatment.

Q3e. Imaging measures in CA prostate

Type: SAQ (10 marks) (mpMRI detail in 2024-Winter Paper 1, Q3d; PSMA PET in 2019-Summer Paper 4, Q3a)

1. Ultrasound

- **TRUS (transrectal ultrasound):** guides biopsy, measures volume (PSA [prostate-specific antigen] density). Cancer is a hypoechoic PZ (peripheral zone) lesion, but **39% are isoechoic**.
- **Also know:** Doppler, contrast and elastography (adjuncts); **micro-ultrasound** (29 MHz; PRI-MUS [Prostate Risk Identification using Micro-Ultrasound] score; close to MRI accuracy).

2. MRI: detection and local staging

- **mpMRI (multiparametric magnetic resonance imaging):**

Sequence	Shows	Cancer
T2-weighted	Anatomy; EPE (extraprostatic extension), SVI (seminal vesicle invasion)	Low signal
DWI (diffusion-weighted imaging) with ADC (apparent diffusion coefficient)	Cellularity	Bright on DWI, dark on ADC
DCE (dynamic contrast-enhanced)	Vascularity	Early enhancement, washout

- **PI-RADS v2.1 (Prostate Imaging Reporting and Data System)** 1–5: DWI dominant in PZ, T2 in TZ (transition zone); DCE only upgrades PZ 3 to 4.
- **PROMIS** (sensitivity 93% vs 48% for TRUS biopsy) and **PRECISION** (more significant, less insignificant cancer): **MRI before biopsy**. Cochrane: sensitivity 0.91, specificity 0.37.
- **Also know:** biparametric MRI; **PI-QUAL** (Prostate Imaging Quality); **PRECISE** (scores change on surveillance); fusion biopsy.

3. Staging by risk group

- **Low or intermediate risk:** no routine CT (computed tomography) or bone scan.
- **High risk:** CT plus **^{99m}Tc-MDP (methylene diphosphonate) bone scan**, or **PSMA (prostate-specific membrane antigen) PET (positron emission tomography)**.
- **proPSMA:** accuracy 92% vs 65% for conventional imaging. Nodal sensitivity only 30–40%, specificity >90%: a negative scan does not exclude micrometastases.

4. Biochemical recurrence

- **PSMA PET** works at low PSA. **Calais:** 56% vs 26% for fluciclovine. **CONDOR:** management changed in 63.9%.
- Choline and fluciclovine PET are older; FDG (fluorodeoxyglucose) PET is poor except in neuroendocrine change.

5. Advanced disease

- PSMA PET selects men for **^{177}Lu-PSMA-617** (VISION); whole-body MRI. No proof yet that PSMA-guided treatment improves survival.

Diagram to draw: prostate zones with a PZ lesion dark on ADC; PSMA PET nodal uptake.

Key points to remember:

- MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
- High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
- PSMA nodal sensitivity 30–40%.

Q3f. Prostatic stents

Type: SAQ (10 marks) (urethral stents also answered in Practice Set 4 Paper 4, Q3f and 2015-Summer Paper 4, Q3d)

Definition

- A device placed in the **prostatic urethra**, between bladder neck and verumontanum, to relieve BOO (bladder outlet obstruction) without resection. Mainly for men **unfit for surgery**.

Classification

Type	Examples	Features
Temporary, first generation	Urospiral, Prostakath (spirals)	Steel or gold-plated coil; not epithelialised
Temporary, second generation	**Memokath** (thermo-expandable nitinol); Allium (covered nitinol)	Expands with warm water, softens with cold water for removal
Temporary polyurethane	Intraurethral catheter, Spanner	Bulbar anchor; sphincter left free
Biodegradable	Polylactic acid stents	Dissolve; trial use
Permanent, epithelialising	**UroLume** (woven mesh)	Urothelium covers the mesh; withdrawn

Indications

- Retention or severe BOO from BPH (benign prostatic hyperplasia) in men unfit for surgery: avoids a long-term catheter.
- Drainage after thermal treatments; trial of de-obstruction.
- **Also know:** DSD (detrusor–sphincter dyssynergia), now abandoned; bulbar stricture.

Contraindications

- Poor detrusor contractility (do urodynamics if in doubt); untreated UTI (urinary tract infection); large median lobe; bladder stones; stricture.

Technique

- Measure prostatic urethral length cystoscopically. Place under local anaesthesia with endoscopic or fluoroscopic guidance. The stent must **end above the external sphincter**, or incontinence follows.

Complications

- **Migration** (commonest with temporary stents); **encrustation**; UTI; irritative symptoms, perineal pain, haematuria.
- **Hyperplastic ingrowth** and restenosis with permanent stents (difficult removal); incontinence if the sphincter is crossed.

Current status

- Most unfit men void after placement, but many stents are later removed.
- **EAU (European Association of Urology):** alternative to catheterisation in men unfit for surgery (weak). Not in the **AUA (American Urological Association) 2023** BPH list.
- **Recent: iTind**: temporary nitinol device left 5–7 days; it remodels the urethra by ischaemic pressure necrosis. AUA 2023 option for 25–75 mL without median lobe. **Optilume BPH** (paclitaxel balloon).

Diagram to draw: sagittal prostate with a Memokath from bladder neck to verumontanum, above the external sphincter.

Key points to remember:

- Temporary (Memokath) vs permanent (UroLume, withdrawn).
- Need detrusor contractility; never cross the sphincter.
- Migration, encrustation, ingrowth; iTind is the modern temporary device.

Rapid Revision Sheet – Practice Set 13 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe role of nephron sparing surgery in management of RCC (LAQ)

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- AUA: RN only if complex + no CKD + new GFR likely >45.
- Preserved parenchyma, not clamp time, governs function.
- EORTC 30904: better function, similar cancer control.

Q2. Desc. factors influencing outcome of ESWL. Discuss complic. & strategies to avoid them. (LAQ)

- Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
- Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
- Slow rate (60–90/min), ramping and good coupling improve breakage and reduce injury.
- Steinstrasse risk rises with burden >2 cm^2; no routine stent.

Q3a. Bicalutamide (SAQ)

- Half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
- Monotherapy raises testosterone; gynaecomastia two-thirds.
- EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
- Withdrawal response 15–30%.

Q3b. Mullerionosis of UB (SAQ)

- Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
- Cyclical haematuria; MRI bright on T1.
- Disease is extravesical: cystoscopy limited, partial cystectomy preferred.

Q3c. Brainstem death (SAQ)

- Six brainstem reflexes absent; apnoea at $PaCO_2$ ≥60 mmHg.
- Four-member Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10.

Q3d. Screening of CA prostate. (SAQ)

- ERSPC: about 20% fall in prostate cancer death; PLCO null from contamination.
- USPSTF Grade C at 55–69.
- MRI-first and active surveillance reduce overdiagnosis and overtreatment.

Q3e. Imaging measures in CA prostate (SAQ)

- MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
- High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
- PSMA nodal sensitivity 30–40%.

Q3f. Prostatic stents (SAQ)

- Temporary (Memokath) vs permanent (UroLume, withdrawn).
- Need detrusor contractility; never cross the sphincter.
- Migration, encrustation, ingrowth; iTind is the modern temporary device.